

THE PETRIDISH



In this issue:

BioBytes

Innovation Insights

Scientists in Spotlight

Study Sphere

Events in Spotlight

Feature Focus

Internship Opportunities

Welcome to the May edition of our newsletter !

As a vibrant hub for biotechnology enthusiasts, “PetriDish” bridges the gap between students, researchers, and industry professionals, offering a window into the latest advancements, groundbreaking research, and inspiring stories from the world of life sciences.

Each edition brings you curated highlights —spotlighting pioneering scientists, showcasing impactful events, and diving deep into topics shaping the future of biotechnology.

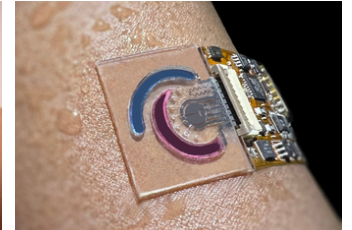
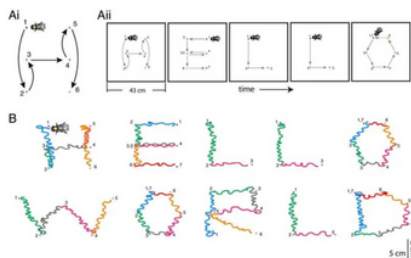
Whether you're passionate about biomedical innovations, eager to explore industry trends, or simply curious about the wonders of biotech, The PetriDish is designed to ignite your curiosity and keep you connected to a community that shares your enthusiasm. Join us as we unravel the fascinating interplay of biology and technology, one edition at a time!

Stay tuned, stay inspired, and let's explore the extraordinary together.

BIOBYTES

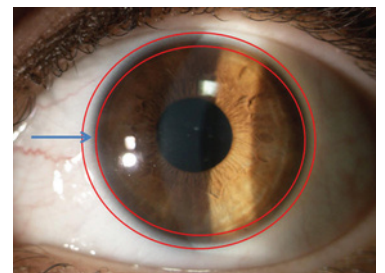
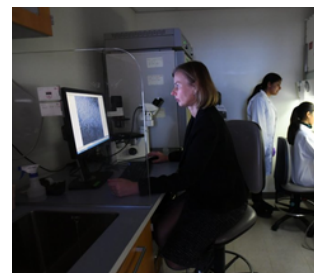
Fruit Flies Become Micro-Robots with a Twist of Light

Scientists at Harvard's Rowland Institute have found a dazzling way to control fruit flies by simply spinning tiny pinwheel-like light patterns! Clockwise light? They turn right. Counterclockwise? They swerve left. This laser-guided technique doesn't harm them but opens a futuristic window into brain mapping, behavior research, and even insect-inspired micro-robotics. Who knew flies could dance to light?



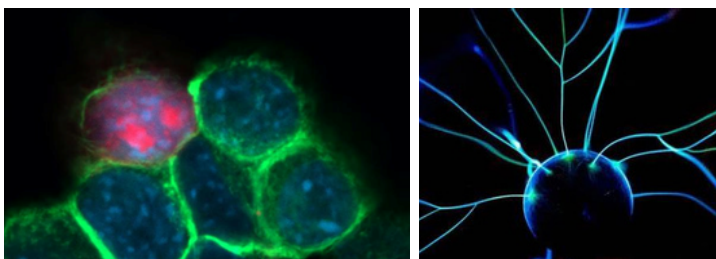
Eyes On the Prize: Restoring Vision with Limbal Cells

Scientists are using cultured limbal stem cells—taken from the edge of a healthy cornea—to repair damaged eyes. These cells can grow and restore vision for people with severe corneal damage, like burns or chemical injuries. The future may hold off-the-shelf eye regeneration kits using your own biology. Sci-fi? Nope, just science.



Impulsonics: Healing With Sound That You Can Feel

A team has developed Impulsonics—a revolutionary tech that uses pulsed ultrasound to push and manipulate cells and tissues inside the body... with pinpoint accuracy and zero cuts. Imagine doing surgery or drug delivery deep inside tissues, using only focused sound waves. It's like having a remote-controlled healing wand inside your body!

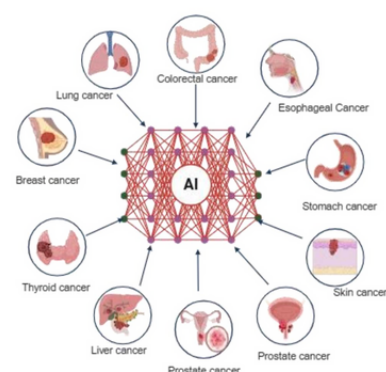


Your Skin is Breathing—And This Device Can Hear It

Researchers from Northwestern have created a non-contact wearable that can actually "sniff" your skin. It reads gas molecules like CO₂ and water vapor coming off your body—without even touching you! It's a game-changer for monitoring wounds, hydration, and even early signs of infection. Think of it like a mini digital nose keeping tabs on your health 24/7.

AI Meets Cancer: Smarter Diagnosis & Targeted Treatment

Artificial Intelligence is quickly becoming a doctor's smartest assistant. A new review highlights how AI is supercharging cancer research—spotting genetic mutations, predicting treatment responses, and analyzing complex data from scans and tissues. From liquid biopsies to radiogenomics, the fusion of machine learning with medicine is rewriting the cancer playbook.



INNOVATION INSIGHTS

De-Extinction & Synthetic Biology: The Dire Wolf Project

Bringing Back an Extinct Giant

De-extinction, the groundbreaking process of bringing back extinct species, marries gene editing, synthetic biology, and genome reconstruction. Colossal Biosciences spearheads this revolution with its ambitious project to restore the dire wolf, a powerful predator that has been extinct for around 10,000 years, renowned for its enormous size and strength.



Engineering the Dire Wolf

Colossal steers clear of cloning from deteriorated fossil DNA, instead piecing together the dire wolf genome through comparison with contemporary relatives such as gray wolves and domesticated dogs. Through the application of CRISPR-Cas9, researchers edit embryos to include dire wolf characteristics, such as greater size, greater bone density, and ancestral behavioral traits.

These attempts have claimed to yield canid pups which are genetic hybrids, showing physical and behavioral traits similar to the ancient dire wolf, but not exact clones.

Ecological and Scientific Impact

The project seeks to recreate ecosystems that were disturbed by the extinction of the dire wolf, introducing a keystone predator for increasing biodiversity. It also offers valuable insights into evolutionary resilience and genetic diversity. Transcending conservation, the synthetic biology methods established here have the potential to improve medicine, including targeted therapies, and agriculture, as in the creation of climate-resilient crops.

Pioneering a New Era

The Dire Wolf Project illustrates how synthetic biology spans past and future. By bringing back an extinct species, Colossal fuels innovation, increases our understanding of nature, and creates new opportunities for solving the world's greatest challenges in health, food security, and environmental sustainability.

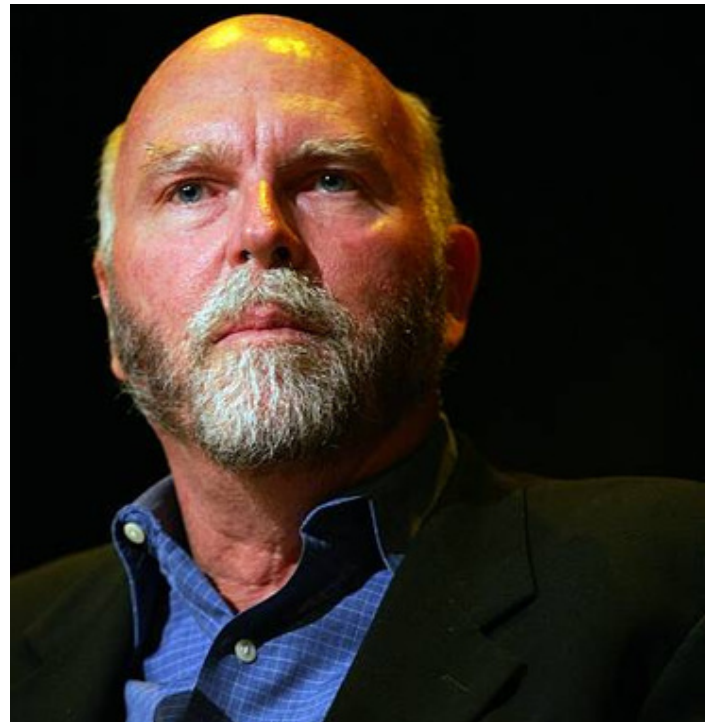


SCIENTIST IN SPOTLIGHT

1) J. Craig Venter: The Maverick Who Redefined Biology

J. Craig Venter, a trailblazing geneticist and entrepreneur, revolutionized biotechnology by leading the first draft of the human genome at Celera Genomics in 2000. Using shotgun sequencing, he surpassed the Human Genome Project, making genomic research faster and more affordable, thus accelerating medical discoveries.

In 2010, Venter achieved a landmark feat by creating *Mycoplasma mycoides* JCVI-syn1.0, the first synthetic life form. His team designed a genome digitally, synthesized it, and transplanted it into a cell, demonstrating that life could be engineered. This breakthrough opened new frontiers in synthetic biology, enabling applications like custom microbes.



Venter's global ocean expeditions uncovered millions of new genes, expanding our understanding of marine biodiversity. Through Human Longevity Inc., he now focuses on decoding aging by sequencing one million human genomes to extend healthy lifespans. His innovative work on DNA-based data storage also aims to preserve humanity's knowledge for millennia.



Awarded the National Medal of Science and the Dan David Prize, Venter embodies disruptive innovation. By transforming biology into an engineerable science, he has redefined what's possible in biotechnology. Venter continues to push boundaries, shaping the future of genomics, synthetic life, and human longevity with his visionary approach.

~Shivam

SCIENTIST IN SPOTLIGHT

2) Dr. Joel Habener: and the discovery of Ozempic

In the 1970s, Dr. Joel Habener's exploration of pancreatic hormones uncovered glucagon-like peptide-1 (GLP-1), a hormone that would transform metabolic disease treatment. His fascination with blood sugar regulation revealed GLP-1's remarkable abilities: stimulating insulin release and suppressing appetite. However, GLP-1's rapid breakdown in the bloodstream posed a challenge.

Collaborating with Dr. Svetlana Mojsov, Habener persevered, finding inspiration in an unexpected source—the Gila monster lizard's venom, which contained a durable GLP-1-like molecule. This discovery led to exenatide, the first of a groundbreaking drug class, paving the way for widely recognized medications like Ozempic.



These therapies have revolutionized diabetes management and facilitated significant weight loss, with patients losing an average of 15% of their body weight, often reducing the need for invasive procedures.

Honored with the 2024 Lasker Award, dubbed “America’s Nobel,” and the 2025 Breakthrough Prize, Habener’s contributions extend beyond diabetes. GLP-1-based treatments are now being explored for Alzheimer’s and heart disease, highlighting the hormone’s vast potential.

Reflecting on his work, Habener notes, “We’re just starting to see GLP-1’s full potential.” His relentless curiosity and persistence transformed a laboratory discovery into a lifeline for millions, redefining medical approaches to metabolic and chronic diseases and cementing his legacy as a pioneer in biotechnology.

~Ojas

STUDY SPHERE

NUCLEOTIDE KNOW-IT-ALL : AI ON GENES



Our bodies function based on instructions in our DNA, organized into genes that code for proteins. Small differences in this DNA, called genetic variants, make us unique and can influence our risk for diseases. Understanding these variants is vital for accurate diagnosis and effective treatment.

Large Language Models (LLMs), a type of AI proficient in understanding language patterns, are now being applied to the "language" of DNA. Trained on vast genomic datasets, LLMs use mechanisms like "self-attention" to identify complex patterns within DNA sequences (A, T, C, G) and predict how variations might lead to disease.

Scientists use LLMs to classify genetic variants through several methods. They combine outputs from multiple models (e.g., GPN-MSA, AlphaMissense) for more comprehensive predictions. Analyzing both DNA and the resulting protein data provides a fuller picture of a variant's impact.

Furthermore, general LLMs can be "fine-tuned" on specific genetic data to become experts in variant classification.

The real-world impact is significant. In heart rhythm disorders, LLMs have drastically improved the classification of previously uncertain variants. They aid in extracting genomic information from cancer reports and help accelerate the diagnosis of rare diseases by analyzing patient data. LLMs are also uncovering insights into the genetic underpinnings of conditions like Alzheimer's disease.

Using LLMs offers key benefits: improved accuracy in variant classification, especially for uncertain variants; significantly faster analysis of vast genetic datasets, speeding up diagnosis; better handling of biological complexity; and the potential to advance personalized medicine. LLMs also streamline the laborious process of reviewing scientific literature, saving valuable time for researchers and clinicians.

~Akash and Samihan

EVENTS IN SPOTLIGHT

1. Industrial Applications of Biotechnology Seminar (January 9th 2025)

Speakers: Francisco Coroado Santos (Oxford) & Preeti Yadav (OmicsLogic)

Dive into AI-powered biotech innovations! From drug discovery to sustainable agriculture, this session highlighted how tech meets biology to solve real-world challenges

2. Invictus'25

BioSoc DTU partnered with Invictus Techfest 2025 to curate a stellar lineup of pre-fest events, setting the stage for innovation and learning.



BioSynergy'25: Computational Biology Panel (February 15th, 2025)

Star Speakers:

1. Dr. Amitansu Pattanaik (Senior Scientist, DRDO, Ministry of Defence)
2. Dr. Dinesh Sharma (Senior Research Fellow at IIT-D, Computational Genomics)
3. Dr. Dinesh Gupta (ICGEB, New Delhi, Translational Bioinformatics)
4. Dr. Ashutosh Shandilya (Researcher at IIT-D, Computational Chemistry and Structural Biology)
5. Dr. Debolina Sarkar (Bioinformatics Scientist at IIF, USA) – via Online Conference Call

Hybrid event alert: Students globally tuned in to learn how algorithms decode diseases and accelerate cures.

Crowd favorite: Dr. Shandilya's 3D molecule simulations!

Lab Rats Quiz 3.0 (February 16th, 2025):

Over 200 participants from DTU, DU, and beyond battled in this high-energy biotech showdown! Teams aced rounds like Disease Dilemma, blending rapid-fire questions and teamwork. Finalists left with sharpened skills and a thirst for biotech innovation.

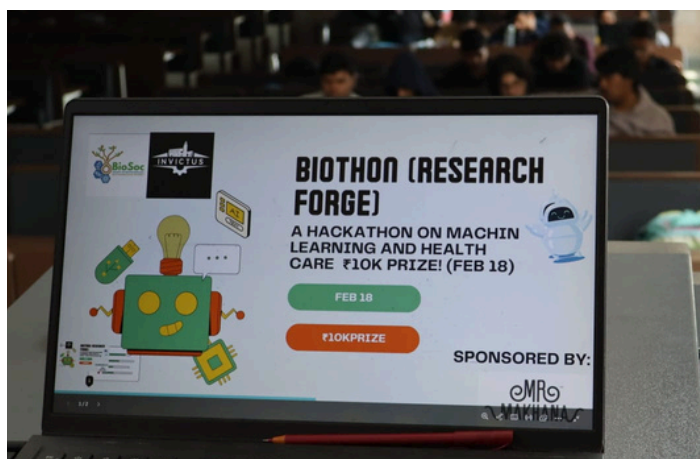
~ Zubinshah and Hunar

EVENTS IN SPOTLIGHT

BioThon 2025: AI x Biotech Revolution (February 18th, 2025):

13 teams. 2 challenges. Infinite ideas.
 Hackathon Heroes: Crunched genomic data for hidden patterns.
 Drug Design Dynamos: Simulated molecules to find the next big cure.
Verdict: AI isn't the future—it's the now of biotech!

From quizzes to hackathons, Invictus'25 ignited curiosity, collaboration, and career dreams. BioSoc DTU continues bridging academia and industry.

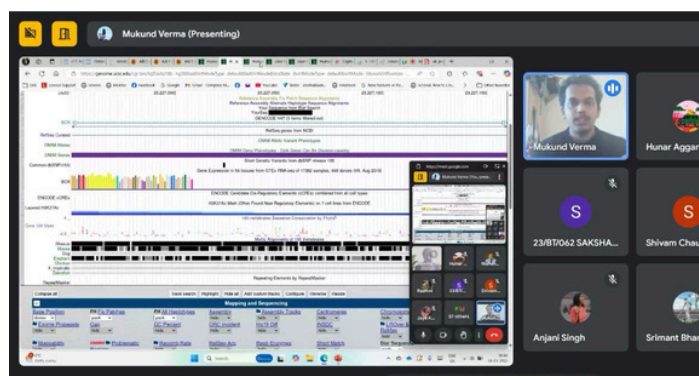


3. Healthcare & Immunology Webinar (March 18th 2025)

Global experts Dr. Nishant Sinha (University of Missouri) and Mukund Verma (Oxford) unpacked advances in

immunology and bioinformatics. Participants gained insights into pandemic preparedness and AI in diagnostics, wrapping up with career Q&As.

Hot topic: "Bioinformatics is the bridge between labs and lives."



4. Workshop on Bioinformatics & Molecular Docking (April 8, 2025):

The workshop provided practical training in molecular docking software and real-case studies, giving the attendees the ability to investigate computational techniques in drug discovery and bioinformatics. Sessions conducted by experts gave a glimpse of the newest tools and methods, helping to gain deeper insights into how bioinformatics fuels innovation in biotechnology. The blended format helped to make it accessible, with both in-person participants at DTU's Pragyan Hall and virtual attendees participating without any hitches.

~ Zubinshah and Hunar

FEATURE FOCUS

Dr. Nishant Sinha on Immunology, Innovation, and AI

Meet Dr. Nishant Sinha, Assistant Research Professor at the University of Missouri, Columbia and Harvard Medical School Research Scholar, who shared some of his cutting-edge biomedical research experience.

1. What inspired your journey into immunology and healthcare research?

My journey into immunology began to expand my understanding of the wound healing process. Immunology has allowed me to study many complex cell types, cytokines, and growth factors to develop new therapies to heal scars. It has also helped me understand the balance between inflammation and regeneration. Through this field, I hope to contribute to therapies that harness the immune system for better healing.

2. What are the most promising innovations in immunology today?

Promising innovations in immunology and healthcare include the development of biologics and cell-based therapies to treat diseases and injuries. These therapies offer personalized treatment options that modulate the immune system more precisely. From monoclonal antibodies to engineered T-cells, such innovations are changing how we approach chronic conditions and repair.

3. How will AI and data science shape the future of immunology and patient care?

AI will allow for the development of novel biomarkers to detect chronic conditions or diseases at an early stage. This will improve patient care and hopefully prevent debilitating conditions from manifesting. By analyzing vast datasets, AI can uncover subtle patterns that might be missed by conventional diagnostic methods.



4. What advice would you give to Indian students aiming for research careers in the U.S.?

The most important part of research is being able to explain and present your data. I would advise to spend the same amount of time practicing your writing and public speaking skills as you spend doing experiments. Clear communication not only helps others understand your work but also opens doors for collaboration, funding opportunities, real-world impact, and career advancement.

5. Can you share a recent exciting project going on in biotech labs?

A recent exciting project in biotech labs involves developing CRISPR-based gene editing therapies to target genetic disorders. Researchers are enhancing the precision and safety of these tools to correct mutations in patient cells. This approach has potential to treat diseases with no cure and is progressing toward clinical trials.

~ Rishabh

INTERNSHIP TIMELINE

Internship opportunities to keep in mind during different times of the year

1. IIIT-Delhi (Indraprastha Institute of Information Technology Delhi)

- Portal to Apply: [IIIT-Delhi Summer Internship Application](#)
- Around February-March

2. IIT Jodhpur (Virtual & Offline)

- Portal to Apply: <https://sites.google.com/iitj.ac.in/summer-internship-2025/home>
- Around March-April.

3. IIT Delhi SERF 2025:

- Apply at academics.iitd.ac.in/srf;
- Around February-March.
- Duration: 8–12 weeks (May–July); stipend ₹500/week + accommodation.

4. IIT Hyderabad SURE 2025:

- Apply at research.iith.ac.in;
- Around February-March.
- Duration: May–July; stipend up to ₹15,000; 250 seats (100 for women), hostel provided.

5. Nestlé India Nesternship 2025:

- Apply at nestle.in/nesternship;
- Duration: 8 weeks from April; includes gamified assessment and multiple business domains.
- Around March.



6. FSSAI Internship 2025:

- Apply at fssai.gov.in;
- Around March-April.
- Duration: 1–3 months; open to UG/PG students in food, law, nutrition, and related fields.

7. Khorana Program for Scholars 2025:

- Apply at iusstf.org;
- Around October.
- Opportunity: 10–12 week summer research internship in top U.S. universities for Indian UG students in biotech, life sciences, and related fields.

8. IISER Pune

- Around February-March.
- Duration: 4–8 weeks
- <http://www3.iiserpune.ac.in/~sspc/>

9. IISc Summer Internship 2025 (Biotech):

- Apply at iisc.ac.in;
- Duration: 8–10 weeks (May–July); work with top researchers in molecular biology, genetics, etc.; stipend and accommodation may vary by lab.
- Around February-March.

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EDITOR'S NOTE



I am filled with immense gratitude as this note marks the end of my editorial journey at BioSoc-DTU. Being the editor has been a highlight of my Master's term here at DTU. This term, my editorship and my senior year are all coming to an end. It is bittersweet, but I'm so grateful.

This newsletter has been a canvas for our ideas and a testament to our passion. I'm deeply grateful for the chance to work with such a dedicated team and for your trust in sharing our society's story. Each issue has captured our collective spirit, and these memories will stay with me always.

Our senior council, who has worked collectively day and night to guide this society, is also retiring. Their efforts have established a solid foundation, and we're excited to hand over the baton to our juniors. They're full of innovation, eager to continue this legacy. To them: the future is yours, and we're so proud.

Thanks for making this journey one to remember. Here's to new beginnings and the lasting spirit of our society.

~Apurva



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